REPORT FOR PDA GROUP PROJECT: SLEEP HEALTH AND LIFESTYLE ANALYSIS USING DATA ANALYTICS

CAMPUS	MALAYSIAN INSTITUTE OF INFORMATION TECHNOLOGY
COURSE NAME	PROGRAMMING FOR DATA ANALYTIC
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1. INTRODUCTION

Sleep is an essential biological process that plays a major role in maintaining physical health, mental well-being, and daily productivity. In modern society, many individuals experience poor sleep quality due to factors such as stress, lifestyle habits, physical inactivity, and unhealthy routines.

This project aims to analyze the relationship between lifestyle factors and sleep health using the Sleep Health and Lifestyle Dataset. By studying variables such as stress level, physical activity, occupation, BMI category, and sleep duration, this project identifies key factors that influence sleep quality and sleep disorders.

2. OBJECTIVES

a) To analyse the relationship between lifestyle factors and sleep quality.
b) To identify factors contributing to sleep disorders.
c) To provide recommendations for improving sleep health.

3. DATA DESCRIPTION

The dataset used in this project is the Sleep Health and Lifestyle Dataset, obtained from Kaggle.

3.1 Dataset Overview

- Number of records: 374
- Number of attributes: 13

3.2 Features Explanation

Feature	Description
Person ID	Unique identifier
Gender	Male / Female
Age	Age of participant
Occupation	Job type
Sleep Duration	Hours of sleep per day
Quality of Sleep	Sleep quality rating
Physical Activity Level	Activity level score
Stress Level	Stress score
BMI Category	Body mass index category
Blood Pressure	Blood pressure reading
Heart Rate	Heart rate value
Daily Steps	Number of steps per day
Sleep Disorder	Type of sleep disorder

Person ID	Gender	Age	Occupatio	Sleep Dura	Quality of	Physical Ac	Stress Leve	BMI Categ	Blood Pres	Heart Rate	Daily Steps	Sleep Disorder
1	Male	27	Software E	6.1	6	42	6	Overweigh	126/83	77	4200	None
2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	None
3	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	None
4	Male	28	Sales Repr	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
5	Male	28	Sales Repr	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
6	Male	28	Software E	5.9	4	30	8	Obese	140/90	85	3000	Insomnia
7	Male	29	Teacher	6.3	6	40	7	Obese	140/90	82	3500	Insomnia
8	Male	29	Doctor	7.8	7	75	6	Normal	120/80	70	8000	None
9	Male	29	Doctor	7.8	7	75	6	Normal	120/80	70	8000	None
10	Male	29	Doctor	7.8	7	75	6	Normal	120/80	70	8000	None
11	Male	29	Doctor	6.1	6	30	8	Normal	120/80	70	8000	None
12	Male	29	Doctor	7.8	7	75	6	Normal	120/80	70	8000	None
13	Male	29	Doctor	6.1	6	30	8	Normal	120/80	70	8000	None
14	Male	29	Doctor	6	6	30	8	Normal	120/80	70	8000	None
15	Male	29	Doctor	6	6	30	8	Normal	120/80	70	8000	None
16	Male	29	Doctor	6	6	30	8	Normal	120/80	70	8000	None
17	Female	29	Nurse	6.5	5	40	7	Normal W	132/87	80	4000	Sleep Apnea
18	Male	29	Doctor	6	6	30	8	Normal	120/80	70	8000	Sleep Apnea
19	Female	29	Nurse	6.5	5	40	7	Normal W	132/87	80	4000	Insomnia
20	Male	30	Doctor	7.6	7	75	6	Normal	120/80	70	8000	None
21	Male	30	Doctor	7.7	7	75	6	Normal	120/80	70	8000	None
22	Male	30	Doctor	7.7	7	75	6	Normal	120/80	70	8000	None
23	Male	30	Doctor	7.7	7	75	6	Normal	120/80	70	8000	None
24	Male	30	Doctor	7.7	7	75	6	Normal	120/80	70	8000	None
25	Male	30	Doctor	7.8	7	75	6	Normal	120/80	70	8000	None
26	Male	30	Doctor	7.9	7	75	6	Normal	120/80	70	8000	None

4. METHODOLOGY

The project was carried out using Python programming in Google Colab with the following steps:

4.1 Data Collection

The dataset was downloaded from Kaggle and imported into Python.

4.2 Importing Libraries

```
[34]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

#import pandas as pd
#Imports the Pandas library.
#Used for:
#reading datasets,
#tables,
#data manipulation.

#pd is just a short nickname.
```

This code imports the required Python libraries. Pandas is used for data manipulation, NumPy for numerical operations, Matplotlib and Seaborn for data visualization.

4.3 Loading Dataset

```
[35]: df = pd.read_excel("Sleep_health_and_lifestyle_dataset.xlsx")

#pd.read_excel()
#Reads Excel file into Python.

#'Sleep_health_and_lifestyle_dataset.xlsx'
#Name of your dataset file.
```

```
[36]: df.head()

#df =

#Stores dataset into variable named:
#df

#df means:
#DataFrame

#A DataFrame is:

#table-like structure,
#rows and columns.
```

This code loads the Sleep Health and Lifestyle dataset into a DataFrame and displays the first five records for initial inspection.

Output:

[36]:

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	1	Male	27	Software Engineer	6.1	6	42	6	Overweight	126/83	77	4200	NaN
1	2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
2	3	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
3	4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
4	5	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea

4.4 Data Inspection

[38]:
```
df.info()

#.info()

#Shows:

#column names,
#data types,
#number of rows,
#missing values.
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 374 entries, 0 to 373
Data columns (total 13 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Person ID                374 non-null    int64
 1   Gender                   374 non-null    object
 2   Age                      374 non-null    int64
 3   Occupation               374 non-null    object
 4   Sleep Duration           374 non-null    float64
 5   Quality of Sleep         374 non-null    int64
 6   Physical Activity Level  374 non-null    int64
 7   Stress Level             374 non-null    int64
 8   BMI Category             374 non-null    object
 9   Blood Pressure           374 non-null    object
 10  Heart Rate               374 non-null    int64
 11  Daily Steps              374 non-null    int64
 12  Sleep Disorder           155 non-null    object
dtypes: float64(1), int64(7), object(5)
memory usage: 38.1+ KB
```

[42]:
```
df.describe()

#.describe()

#Shows statistics such as:

#mean,
#minimum,
#maximum,
#standard deviation.
```

[42]:

	Person ID	Age	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	Heart Rate	Daily Steps
count	374.000000	374.000000	374.000000	374.000000	374.000000	374.000000	374.000000	374.000000
mean	187.500000	42.184492	7.132086	7.312834	59.171123	5.385027	70.165775	6816.844920
std	108.108742	8.673133	0.795657	1.196956	20.830804	1.774526	4.135676	1617.915679
min	1.000000	27.000000	5.800000	4.000000	30.000000	3.000000	65.000000	3000.000000
25%	94.250000	35.250000	6.400000	6.000000	45.000000	4.000000	68.000000	5600.000000
50%	187.500000	43.000000	7.200000	7.000000	60.000000	5.000000	70.000000	7000.000000
75%	280.750000	50.000000	7.800000	8.000000	75.000000	7.000000	72.000000	8000.000000
max	374.000000	59.000000	8.500000	9.000000	90.000000	8.000000	86.000000	10000.000000

The code is used to understand the structure of the dataset, including data types, number of records, and statistical summaries of numerical attributes.

4.5 Data Cleaning

[41]:
```
df.isnull().sum()

#.isnull()

#Checks:

#missing values (NaN).

#Returns:

#True/False.
#.sum()

#Counts:

#total missing values.
```

```
[41]: Person ID                  0
      Gender                     0
      Age                        0
      Occupation                 0
      Sleep Duration             0
      Quality of Sleep           0
      Physical Activity Level    0
      Stress Level               0
      BMI Category               0
      Blood Pressure             0
      Heart Rate                 0
      Daily Steps                0
      Sleep Disorder           219
      dtype: int64
```

```
[43]: df.drop_duplicates(inplace=True)

      #.drop_duplicates()

      #Removes repeated rows.

      #inplace=True

      #Means:

      #modify original dataset directly.

      #Without it:

      #changes are temporary.
```

Missing values and duplicate records were checked and removed to improve data quality and ensure accurate analysis.

4.6 Exploratory Data Analysis

EDA was performed to understand data distributions and relationships between variables.

The following analyses were conducted:

- Gender Distribution
- Sleep Duration Distribution
- Stress Level vs Sleep Quality
- Physical Activity vs Sleep Quality
- Occupation vs Sleep Duration
- Sleep Disorder Analysis
- Correlation Analysis

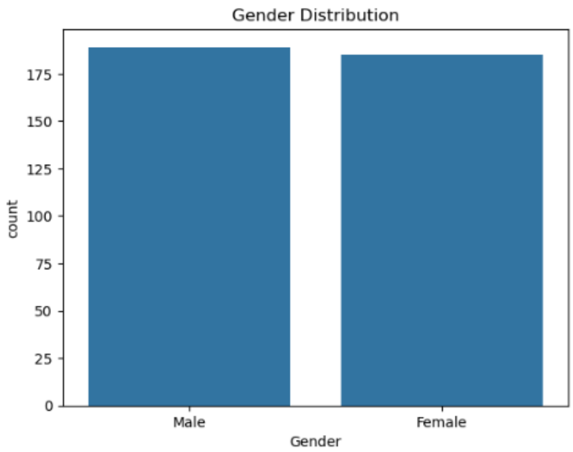
4.7 Tools Used

EDA was performed to understand patterns and relationships between variables using statistical methods and visualization tools.

5. RESULT AND DISCUSSION

5.1 Gender Distribution

```python
sns.countplot(x='Gender', data=df)

plt.title('Gender Distribution')
plt.show()

#sns.countplot()

#Creates:
#count graph for categories.

#x='Gender'
#Uses:
#Gender column on X-axis.

#data=df
#Uses:
#dataset named df.

#plt.title()
#Adds graph title.

#plt.show()
#Displays graph.
```



Explanation

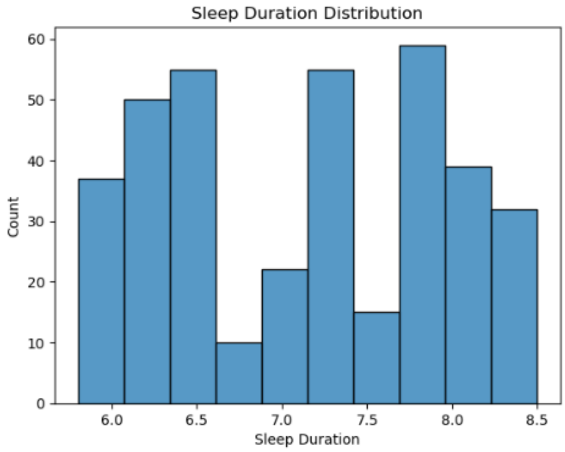
This code generates a count plot showing the distribution of male and female participants in the dataset.

Findings

The dataset contains a balanced number of male and female participants.

5.2 Sleep Duration Analysis

```
[48]: sns.histplot(df['Sleep Duration'], bins=10)

plt.title('Sleep Duration Distribution')
plt.xlabel('Sleep Duration')
plt.show()

#sns.histplot()
#Creates histogram.
#Used for:
#numerical distributions.

#df['Sleep Duration']
#Selects:
#Sleep Duration column.

#bins=10
#Divides graph into:
#10 intervals/groups.
```



Explanation

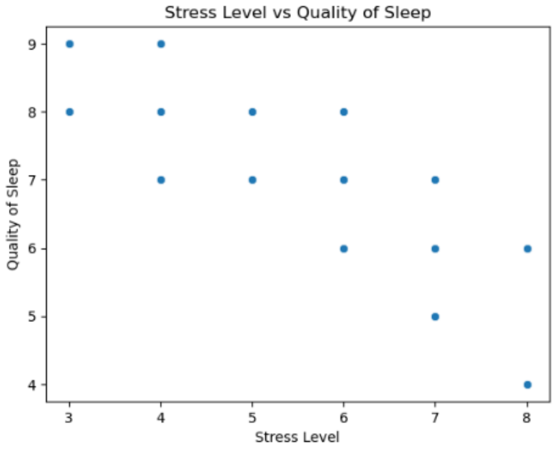
The histogram visualizes the distribution of sleep duration among participants.

Findings

Most participants sleep between 6 and 8 hours per day.

5.3 Stress Level vs Sleep Quality

```
[49]: sns.scatterplot(
    x='Stress Level',
    y='Quality of Sleep',
    data=df
)

plt.title('Stress Level vs Quality of Sleep')
plt.show()

#sns.scatterplot()
#Creates scatterplot.
#Used for:
#relationship between two variables.

#x='Stress Level'
#Stress Level on X-axis.

#y='Quality of Sleep'
#Sleep quality on Y-axis.
```



Explanation

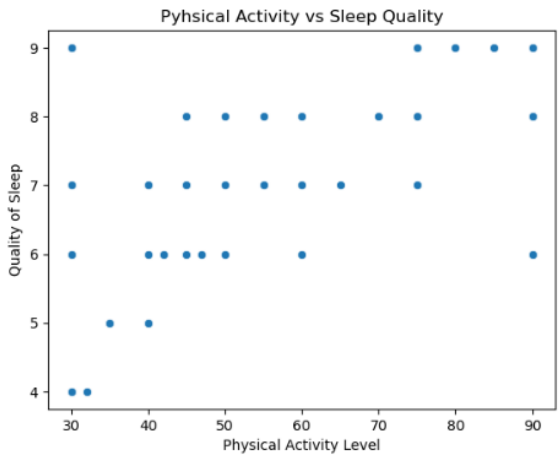
The scatter plot is used to examine the relationship between stress level and sleep quality.

Findings

Higher stress levels are generally associated with lower sleep quality.

5.4 Physical Activity vs Sleep Quality

```
[51]: sns.scatterplot(
    x='Physical Activity Level',
    y='Quality of Sleep',
    data=df
)

plt.title('Pyhsical Activity vs Sleep Quality')
plt.show()

#Another scatterplot.

#Purpose:
#analyze relationship between:
#exercise,
#sleep quality.
```



Explanation

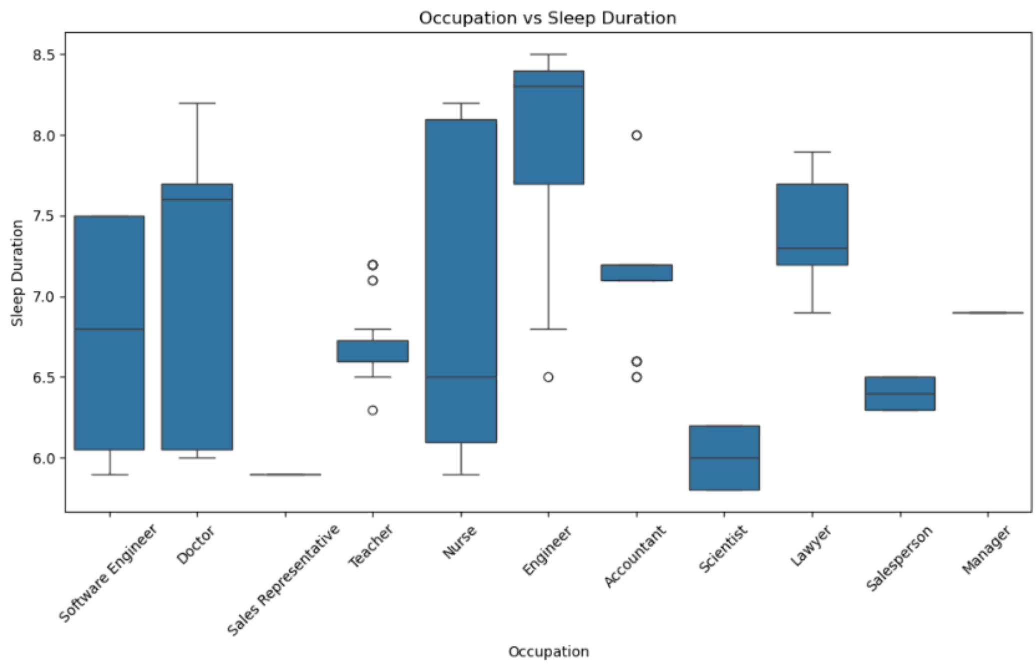
This visualization analyzes whether physical activity influences sleep quality.

Findings

Participants with higher physical activity levels tend to have better sleep quality.

5.5 Occupation vs Sleep Duration

```
[50]: plt.figure(figsize=(12,6))

sns.boxplot(
    x='Occupation',
    y='Sleep Duration',
    data=df
)

plt.xticks(rotation=45)
plt.title('Occupation vs Sleep Duration')
plt.show()

#sns.boxplot()
#Creates boxplot.

#WHAT BOXPLOT SHOWS
#median,
#quartiles,
#outliers,
#distribution.

#x='Occupation'
#Occupation categories.

#y='Sleep Duration'
#Sleep hours.
```



Explanation

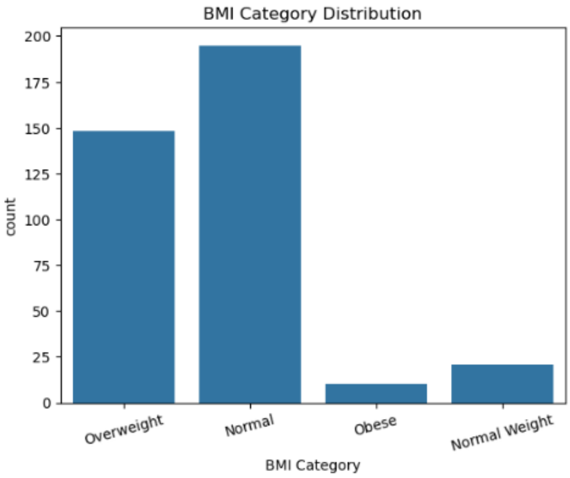
This boxplot compares sleep duration across different occupations. It helps identify whether certain professions tend to sleep more or less than others.

Findings

The graph shows that sleep duration varies among occupations. Some occupations, such as healthcare workers or professionals with demanding schedules, tend to have lower sleep duration compared to others. This suggests that job responsibilities and work-related stress may affect sleeping habits.

5.6 BMI Category Distribution

[52]:
```
sns.countplot(x='BMI Category', data=df)

plt.title('BMI Category Distribution')
plt.xticks(rotation=15)
plt.show()

#Creates countplot for:
#BMI categories.

#Purpose:
#determine health category distribution.
```



Explanation

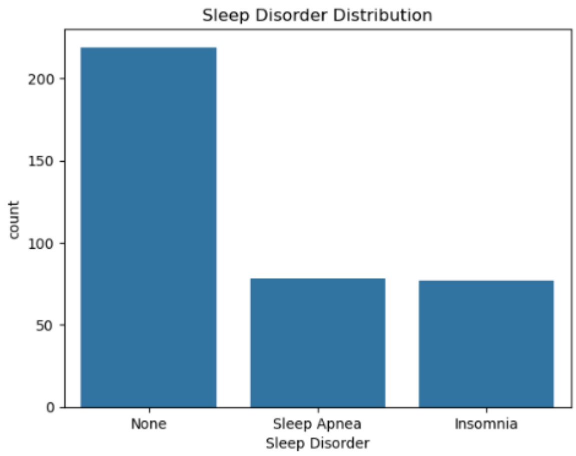
This count plot shows the distribution of participants across different BMI categories, such as Normal, Overweight, and Obese. BMI (Body Mass Index) is an important health indicator that can influence sleep quality and overall well-being.

Findings

The graph shows that most participants fall into the Normal and Overweight BMI categories. Fewer participants are classified as Obese. This suggests that most individuals have a relatively healthy or slightly elevated body weight.

5.7 Sleep Disorder Analysis

```python
sns.countplot(x='Sleep Disorder', data=df)

plt.title('Sleep Disorder Distribution')
plt.show()

#Counts:
#insomnia,
#sleep apnea,
#none.

#Purpose:
#analyze disorder prevalence.
```



Explanation

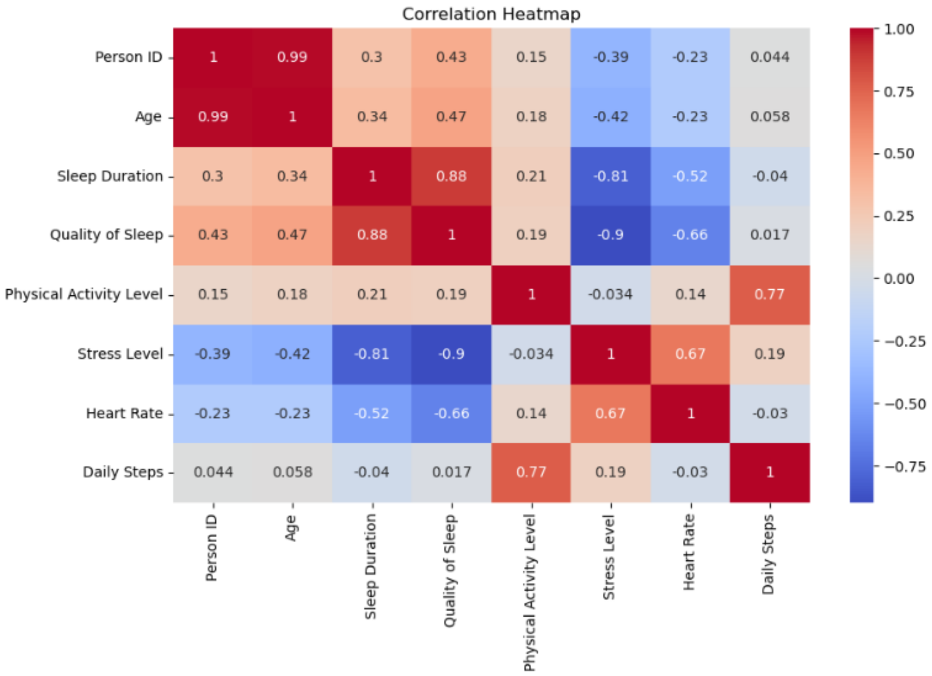
The count plot displays the frequency of different sleep disorders among participants.

Findings

Insomnia and Sleep Apnea are the most common sleep disorders observed.

5.8 Correlation Heatmap

```python
plt.figure(figsize=(10,6))

sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap='coolwarm'
)

plt.title('Correlation Heatmap')
plt.show()

#plt.figure(figsize=(10,6))
#Sets graph size:
#width = 10
#height = 6

#df.corr()
#Calculates correlations.

#CORRELATION
#Measures relationship strength:
#+1 → strong positive
#-1 → strong negative
#0 → no relationship

#numeric_only=True
#Uses only numerical columns.
#Because:
#correlation cannot use text data.
#annot=True
#Displays numbers inside heatmap.

#cmap='coolwarm'
#Sets color theme.
#Usually:
#red = positive
#blue = negative
```



Explanation

A correlation heatmap is used to identify relationships between numerical variables.

Findings

Stress level shows a negative relationship with sleep quality, while physical activity has a positive relationship.

6. FINDINGS

From the analysis, the following findings were obtained:

- High stress reduces sleep quality
- Physical activity improves sleep quality
- Occupation affects sleep duration
- BMI and lifestyle influence sleep disorders

7. GITHUB REPOSITORY

LINK: https://github.com/MeiiRaah/IIB40303GroupProject.git

8. CONCLUSION

This project successfully analysed the Sleep Health and Lifestyle Dataset to understand the factors affecting sleep quality and sleep disorders. The results show that stress level, physical activity, occupation, and BMI category significantly influence sleep health.

Individuals with lower stress levels and higher physical activity tend to experience better sleep quality. Therefore, maintaining a healthy lifestyle and managing stress are important for improving sleep health.

9. APPENDIX

YOUTUBE LINK: **https://youtu.be/hsPfVmntexU?si=ymnQk3pxk3tseTs_**